

504MAN Resource Management

Capacity Management & MRP

Week 3 – Session 11

Don't forget to tap in

Our responsibilities:

- Understand why students aren't attending.
- Provide support to help you succeed.
- Be compliant with regulations and educational policies.
- Report students who are not attending to UKVI.

Your responsibilities:

- Use your own ID card to register your attendance.
- Be familiar with and follow institutional policies and legal regulations.

- !! **Do NOT tap in on behalf of other students.** This can result in a violation of visa conditions and/or institutional policy and procedures.
- !! **Do NOT give your student ID card to someone else** to let them into the library or other campus buildings as this is an offence under institutional regulations.



Extensions & Deferrals – What You Need to Know

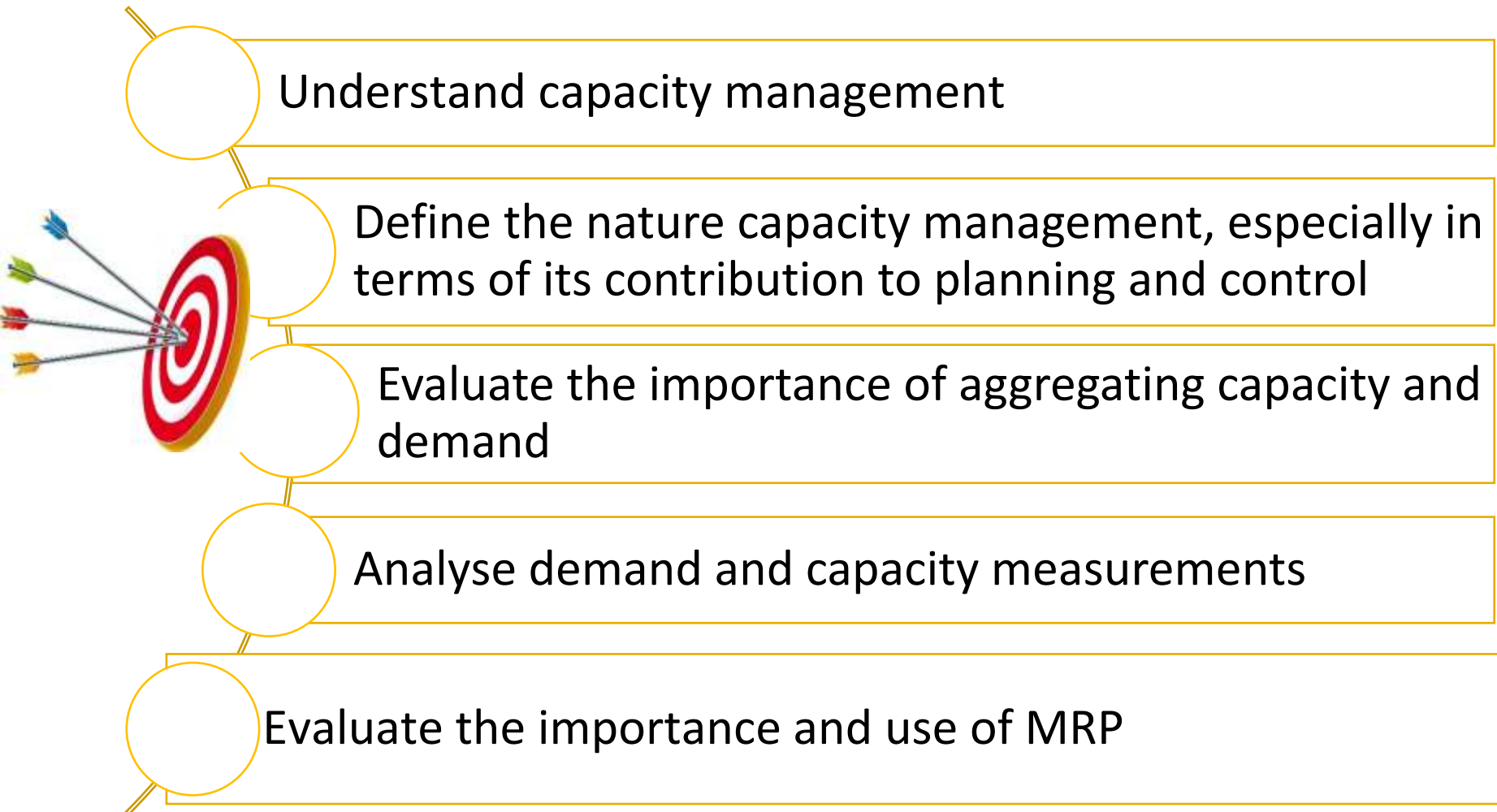
-  Deferring may delay course progression or graduation.
- Tier 4/Student Visa holders: Deferring may affect Post Study Visa eligibility.
- For visa advice, contact: visahelp.io@coventry.ac.uk
-  Use one form per semester for multiple components/modules.
- Do not submit multiple forms.
- Form link: <https://livecoventryac.sharepoint.com/sites/students-registry-extensions-deferrals/SitePages/Extension---Deferral-Form.aspx>
-  Before submitting:
- Engage with your tutor and library academic writing support.
- Where required, seek support from our Welfare Team through Spectrum Life
- For help with form completion, contact the Student Success Coaches at Main Reception.

Intended Module Learning Outcomes

1. Analyse the importance of effective and efficient use of resources in organisations
2. Analyse principles which underpin effective procurement of resources
3. Investigate and plan the resources required to achieve an organisations aim and objectives
4. Analyse the procedures and practices that ensures resources are maximised and remedial actions taken where appropriate
5. Evaluate and appraise the use of resources against organisational plans
6. Analyse principles that underpin effective supply chain management

Session Outcomes

At the end of this session learners should be able to:



Capacity

- Capacity is maximum level of value-added activity over a period of time that process or operation can achieve under normal operating conditions

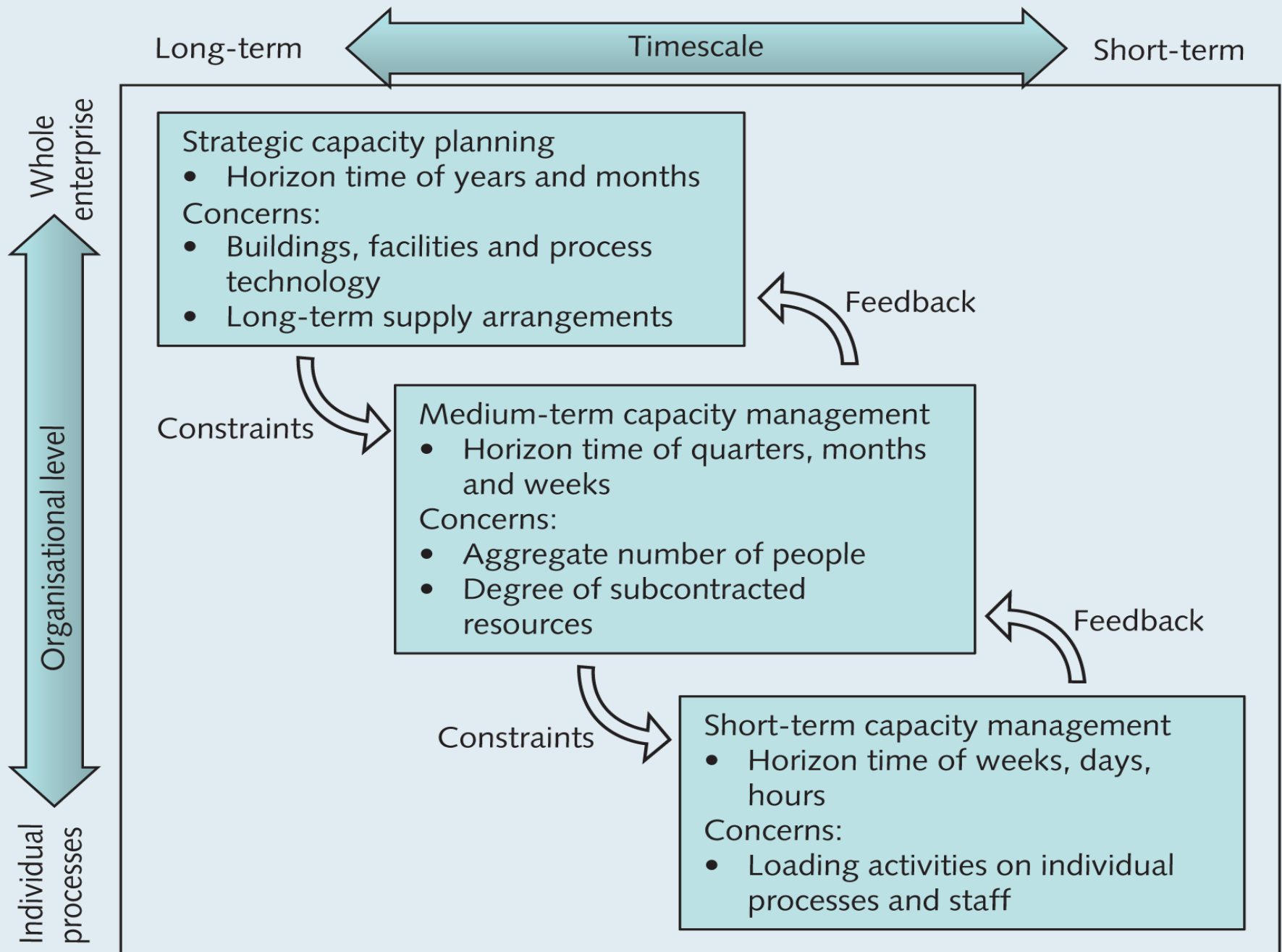
Capacity Management

- Capacity management is the activity of understanding the nature of demand for goods and services, and effectively planning and controlling capacity in the short/medium/long term

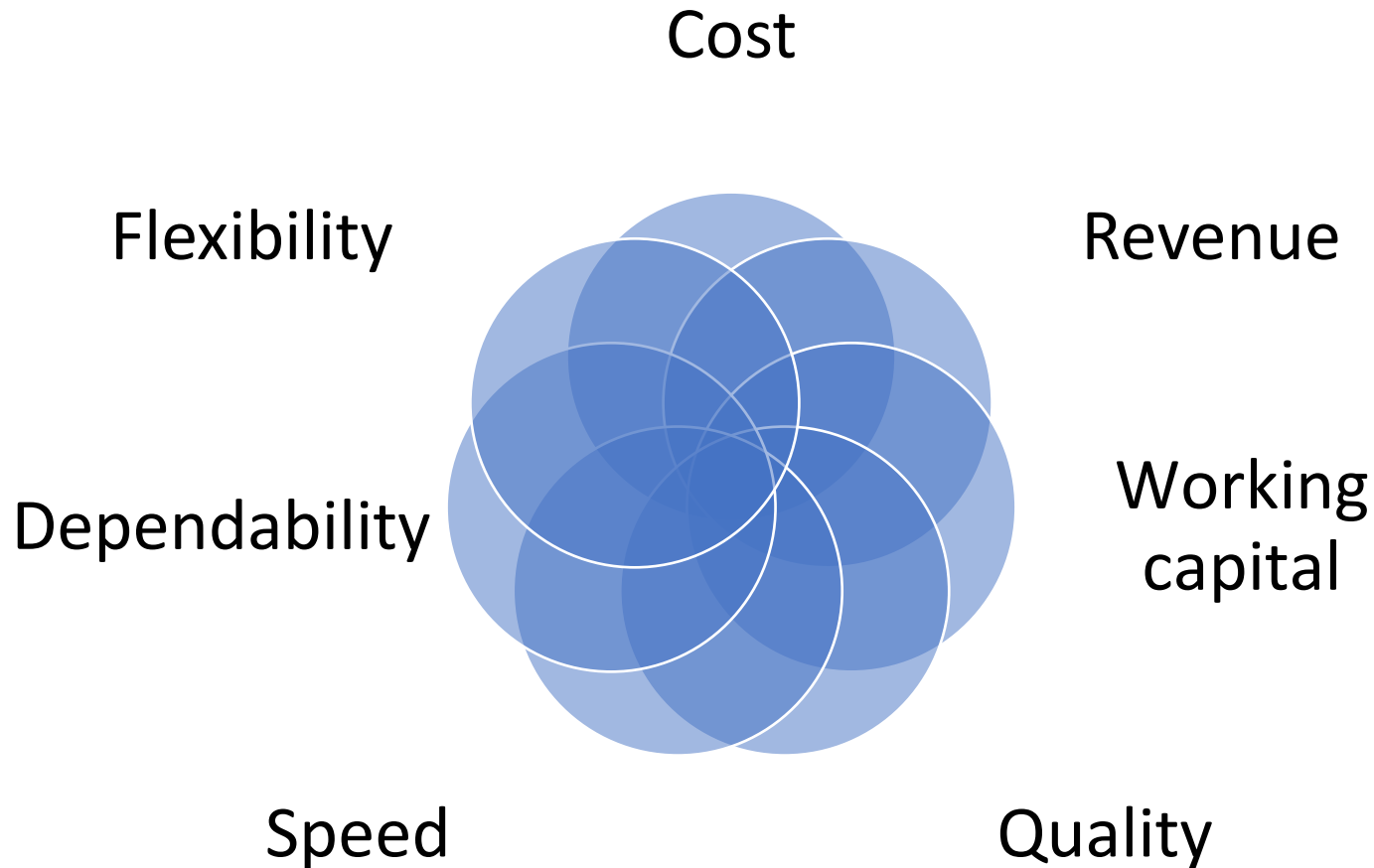
What is Capacity Management in Business Operations?

- Watch the video below.

https://www.youtube.com/watch?v=7u5z21B_FSM



The Objectives of Capacity Management



Measure aggregate demand (total demand) and capacity



Understand the changes to demand and capacity



Determine the operation's base level of capacity



Identify and select methods of coping with mismatches



Understand the consequences of different capacity decisions

Aggregate Demand and Capacity

- The important characteristic of capacity managements is that it is concerned with setting capacity level over a range of terms (aggregated terms)
- The objective of aggregate planning is usually to meet forecast demand while minimizing cost over the planning period

Aggregate Planning Strategies



Should inventories be used to absorb changes in demand?

Should changes be accommodated by varying the size of the workforce?

Should part-timers be used, or should overtime or idle time absorb fluctuations?

Should subcontractors be used and maintain a stable workforce?

Should prices or other factors be changed to influence demand?

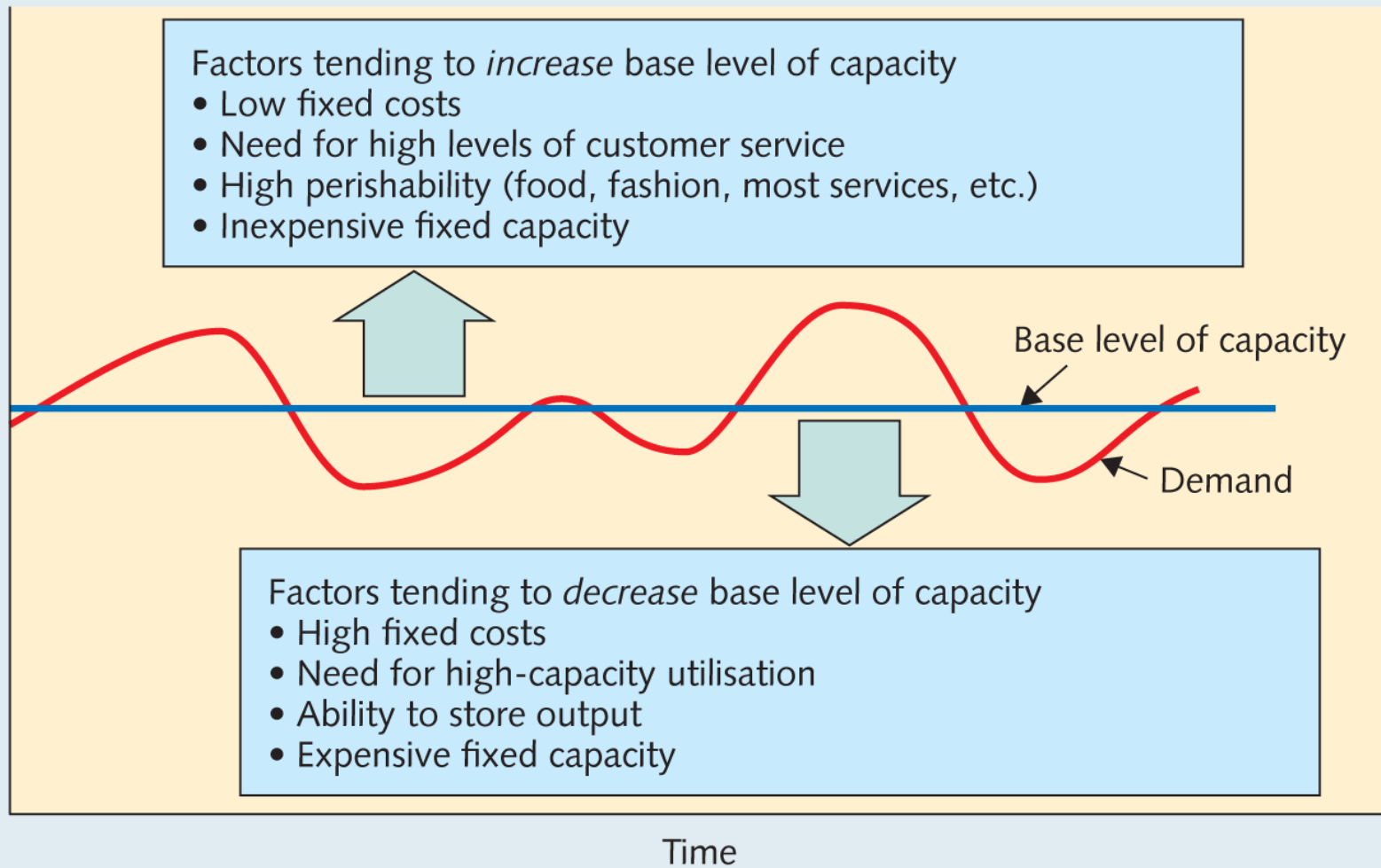
Setting Base Capacity

The base level of capacity in any operations is influenced by particularly

The relative importance of the operation performance objectives

The perishability of the operation's outputs

The degree of variability in supply or demand



- The base level of capacity should reflect the relative importance of the operation's performance objectives

Mini Case Study: Read and Discuss



Activity

10 minutes

OM in Action

Matching Airline Capacity to Demand

Airlines constantly struggle to control their capital expenditures and to adapt to unstable demand patterns.

Southwest and Lufthansa have each taken their own approach to increasing capacity while holding down capital investment. To manage capacity constraints on the cheap, Southwest squeezes seven flight segments out of its typical plane schedule per day—one more than most competitors. Its operations personnel find that quick ground turnaround, long a Southwest strength, is a key to this capital-saving technique.

Lufthansa has cut hundreds of millions of dollars in new jet purchases by squashing rows of seats 2 inches closer together. On the A320, for example, Lufthansa added two rows of seats, giving the plane 174 seats instead of 162. For its European fleet, this is the equivalent of having 12 more Airbus A320 jets. But Lufthansa will tell you that squeezing in more seats is not quite as bad as it sounds, as the new generation of ultra-thin seats provides passengers with more leg room. Using a strong mesh, similar to that in fancy office chairs

(instead of inches of foam padding), and moving magazine pockets to the top of seat backs, there is actually more knee room than with the old chairs.

Unstable demands in the airline industry provide another capacity challenge. Seasonal patterns (e.g., fewer people fly in the winter), compounded by spikes in demand during major holidays and summer vacations, play havoc with efficient use of capacity. Airlines attack costly seasonality in several ways. First, they schedule more planes for maintenance and renovations during slow winter months, curtailing winter capacity; second, they seek out contra-seasonal routes. And when capacity is substantially above demand, placing planes in storage (as shown in the photo) may be the most economical answer.

Airlines also use revenue management to maximize per-seat pricing of available capacity, regardless of current demand patterns.

Sources: The Wall Street Journal (February 29, 2012) and (October 6, 2011).

Short-term outlook for volume

		Decreasing below current capacity	Level with current capacity	Increasing above current capacity
Long-term outlook for volume	Decreasing below current capacity	Reduce capacity (semi) permanently. For example, reduce staffing levels; reduce supply agreements.	Plan to reduce capacity (semi) permanently. For example, freeze recruitment; modify supply agreements.	Increase capacity temporarily. For example, increase working hours, and/or hire temporary staff; modify supply agreements.
	Level with current capacity	Reduce capacity temporarily. For example, reduce staff working hours; modify supply agreements.	Maintain capacity at current level.	Increase capacity temporarily. For example, increase working hours, and/or hire temporary staff; modify supply agreements.
	Increasing above current capacity	Reduce capacity temporarily. For example, reduce staff working hours, but plan to recruit; modify supply agreements.	Plan to increase capacity above current level; plan to increase supply agreements.	Increase capacity (semi) permanently. For example, hire staff; increase supply agreements.

- Capacity management strategies are partly dependent on the long and short term outlook for volumes



**LET'S
DISCUSS**

5 mins

my stock photo

Materials Requirements Planning



What is Materials Requirement Planning?

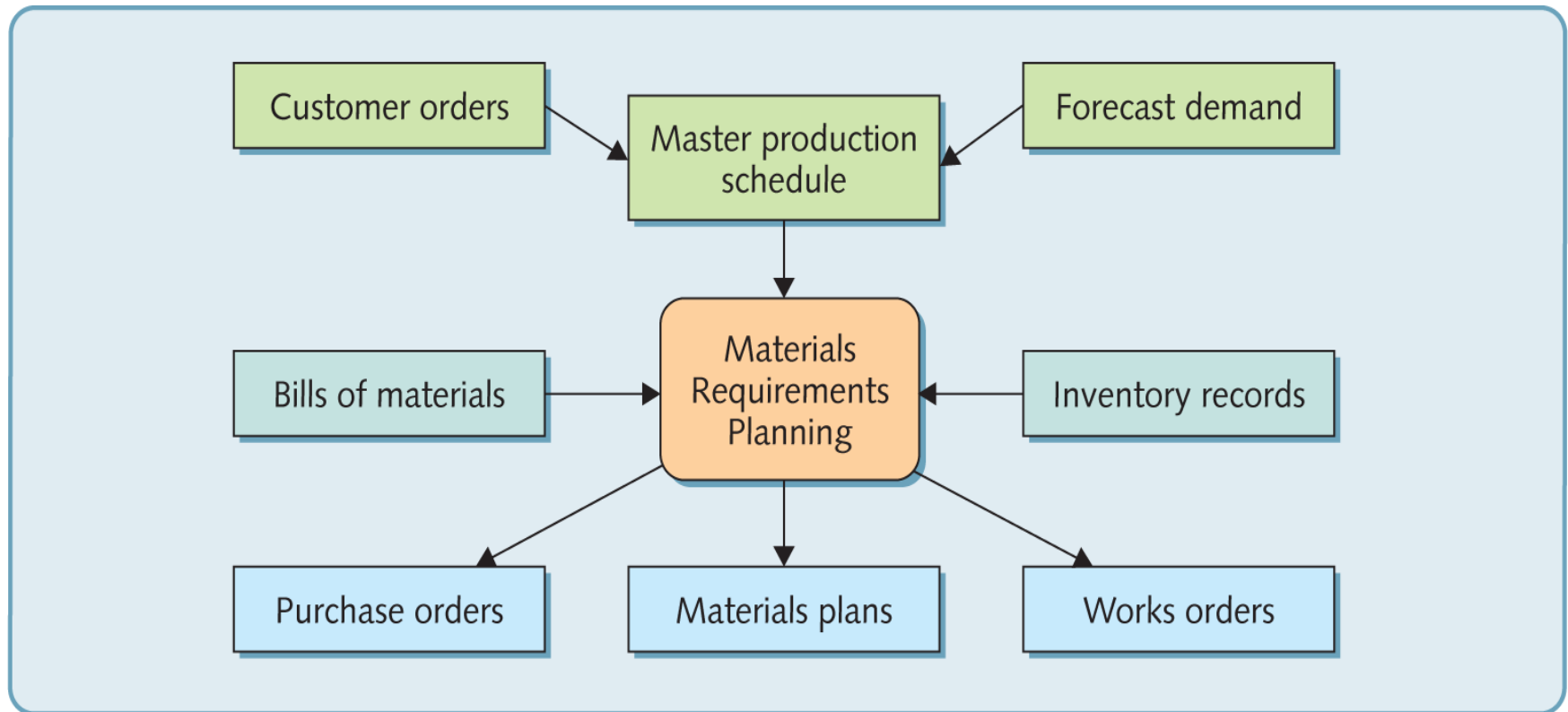
- Watch the video below.

<https://www.youtube.com/watch?v=eHjjRXc3MR8>

Materials Requirements Planning (MRP)

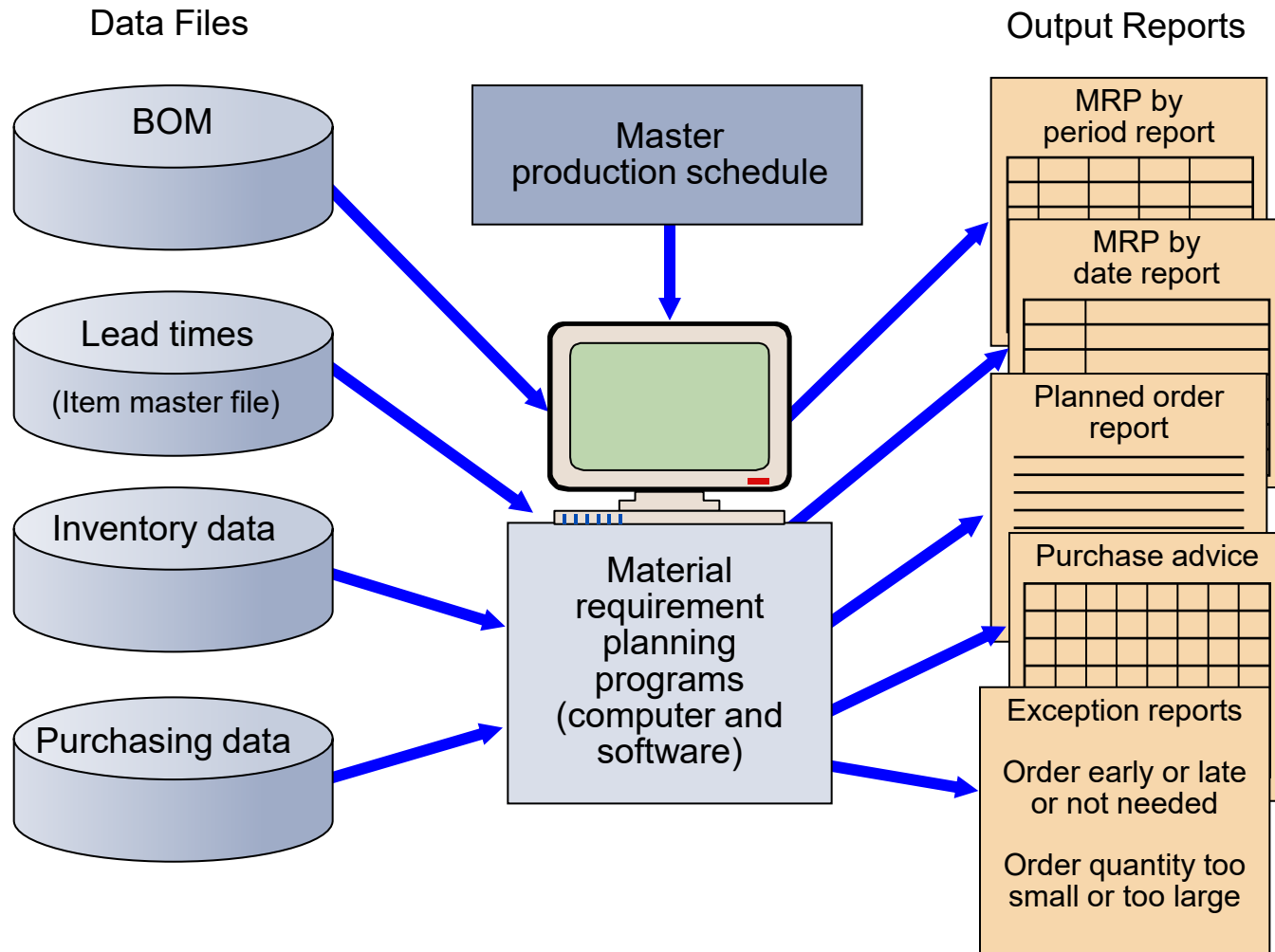
- is an approach to calculating how many parts or materials of particular types are required and what times they are required
- this requires data files, which, when the MRP program is run, can be checked and updated

Materials requirements planning (MRP) schematic



The first inputs to MRP are customer orders and forecast demand
All other requirements are derived from, and dependent on, demand data

MRP Structure



MRP Limitations

- MRP does not do detailed scheduling—it plans
- Works best in product-focused, repetitive environments
- Requires fixed lead time and infinite size time buckets
 - MRP will tell you that a job needs to be completed on a certain week or day but does not tell you that Job X needs to run on Machine A at 10:30 a.m. and be completed by 11:30 a.m. so that Job X can then run on Machine B
 - The buckets are time units, usually one week. MRP puts work into these buckets without regard to capacity.
 - Consequently, MRP is considered an infinite scheduling technique

Benefits of MRP

- Better response to customer orders
- Faster response to market changes
- Improved utilization of facilities and labour
- Reduced inventory levels

These benefits are the result of a strategic decision to use a dependent inventory scheduling system

Material requirements planning (MRP) is a dependent demand technique that uses a bill-of-material, inventory, expected receipts, and a master production schedule to determine material requirements.

Master Production Schedule (MPS)

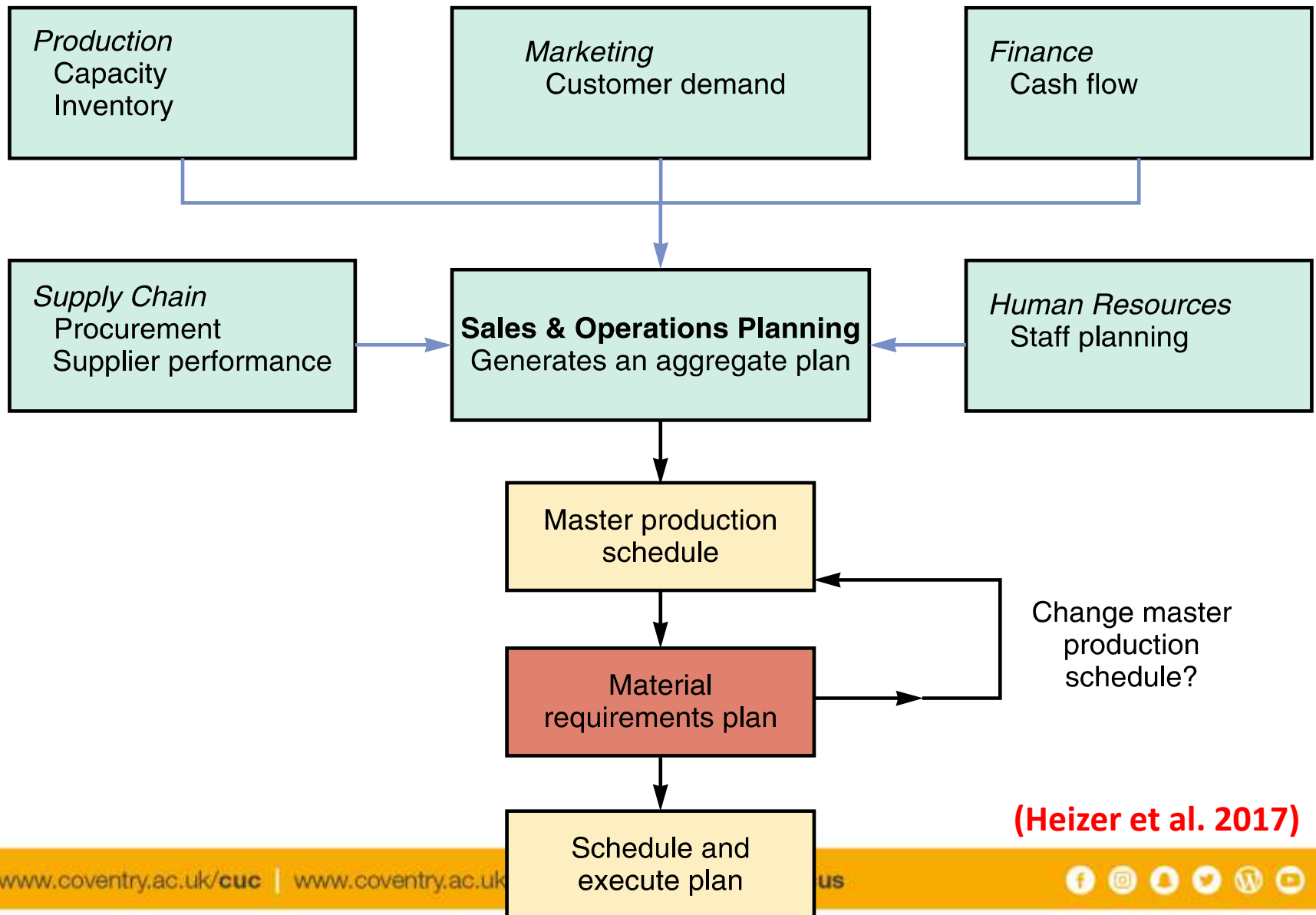
- MPS forms of the main input to MRP and contains a statement of the volume and timing of the end products to be made
- Specifies what is to be made and when
- Must be in accordance with the aggregate production plan
- Inputs from financial plans, customer demand, engineering, labor availability, inventory fluctuations, supplier performance
- As the process moves from planning to execution, each step must be tested for feasibility

Master Production Schedule (MPS)

[contd]

- MPS is established in terms of specific products, it disaggregates the aggregate plan
- Schedule must be followed for a reasonable length of time
- The MPS is quite often fixed or frozen in the near-term part of the plan
- The MPS is a rolling schedule
- The MPS is a statement of what is to be produced, not a forecast of demand

The Planning Process



Accurate Inventory Records

- Accurate inventory records are absolutely required for MRP (or any dependent demand system) to operate correctly
- If the firm does not exceed 99% record accuracy, then material requirements planning will not work.



Accurate Inventory Records

Why Is Accurate Inventory Record Important?

- Investors want to know that the book value is accurate.
- Lenders who loan money with inventory as collateral want to protect their loan.
- Taxation often depends on inventory value. Overpayment of taxes reduces profits and underpayment incurs penalties.
- Poor accuracy begets more inventory and requires more capital. Inventory is often the largest consumer of capital for an enterprise.
- Stockouts interrupt production and create delivery delays.
- People waste hours looking for misplaced or missing items.
- When stock-outs are frequent, inventory rises to compensate. This unnecessary inventory requires space and capital.
- Inventory turnover reflects overall manufacturing efficiency

Team activity – 30 minutes

- In teams of 3 or 4, read the article on Aula - “Revamping fashion sourcing: Speed and flexibility to the fore”.
- Using flip chart paper, create a mind map highlighting key aspects of the article. You should aim to find examples from real organisations who have gone through some of the issues highlighted in the article.
- Present your mind map to the rest of the class.

Traffic Light *it*

Where is your learning at?



Activity

5 minutes



Red = I have misunderstood some of the topics. (write down 2 questions that you need help with)

Amber = I have understood most of the topics so far (write down 1 question and 1 key theme you have understood)

Green = I have understood the topics so far (write down 2 key themes from the lesson to show your understanding)

Session Reference List

- Greasley, A. (2013) Operations Management. 3rd edn. John Wiley & Sons
- Heizer, J., Render, Barry, author, & Munson, Chuck, author. (2017). Principles of operations management sustainability and supply chain management (Tenth edition.; Global ed.).
- Hill, A. and Hill, T. (2018) Essential operations management. Second edition. London: Palgrave
- Slack, N., Brandon-Jones, Alistair, author, & Johnston, Robert, author. (2013). *Operations management* (Seventh ed.). Harlow, England: Pearson
- Slack, N. et al. (2015) Operations and process management: principles and practice for strategic impact. Fourth edition. Harlow, England: Pearson
- Slack, N. and Brandon-Jones, A. (2018) Operations and process management: principles and practice for strategic impact. Fifth edition. Harlow, England: Pearson

Reading



Essential Reading

- Torrington, D., Hall, L., Atkinson, C. (2025). *Human Resource Management* (12th edition). Pearson
- Torrington, D., Hall, L., Atkinson, C. (2020). *Human Resource Management* (11th edition). Pearson

Recommended Reading

- Noe, R.A., Hollenbeck, J.R., Gerhart, B.A., & Wright, P.M. (2024). *Fundamentals of human resource management* (Ninth edition). McGraw Hill.
- Noe, R.A., Hollenbeck, J.R., Gerhart, B.A., & Wright, P.M. (2022). *Human resource management: gaining a competitive advantage* (Thirteenth edition). McGraw Hill.
- Heizer, J., Render, B., & Munson, C. (2024). *Principles of Operations Management: Sustainability and Supply Chain Management* (Twelfth edition). Pearson.
- Slack, N., & Brandon-Jones, A. (2024). *Operations and process management* (7th edition). Pearson.
- Slack, N., & Brandon-Jones, A. (2018). *Operations and process management principles and practice for strategic impact* (Fifth edition). Pearson.
- Hill, A., & Hill, T. (2011). *Essential operations management* (Second edition). Palgrave.

Additional Reading

- Bratton, J., & Bratton, A., (2022) *Human Resource Management: A critical approach*. 7th edn. London UK: Bloomsbury Publishing Plc
- **CMI Website : Management Direct**